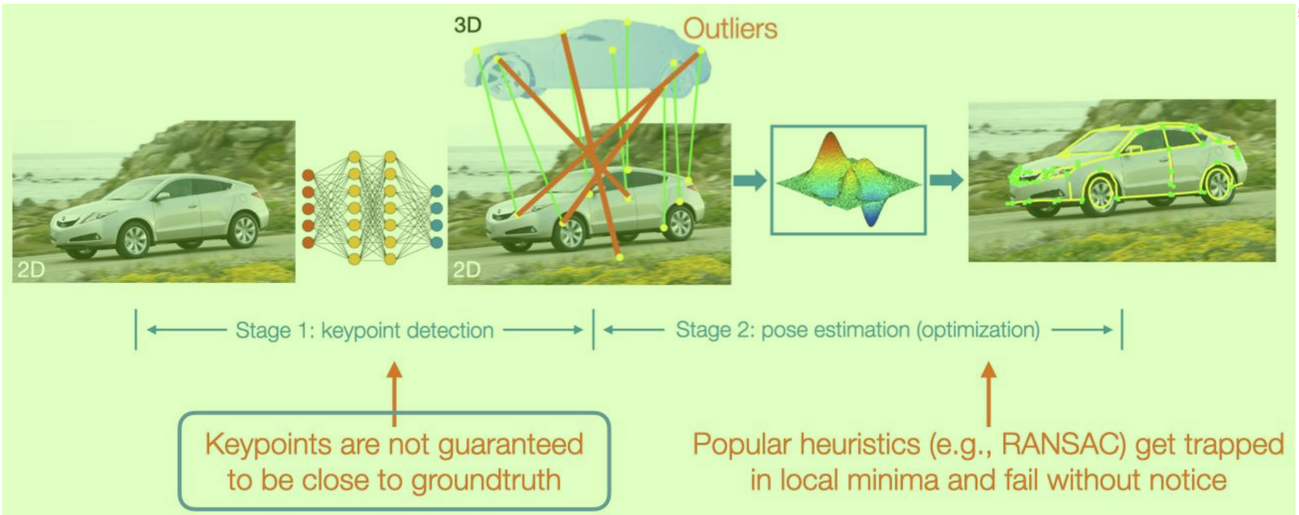
Conformal Semantic Keypoint Detection with Statistical Guarantees [1]

Heng Yang and Marco Pavone [2]

Workshop on Robot Learning: Trustworthy Robotics [3]

Object Pose Estimation: why untrustworthy? [4]

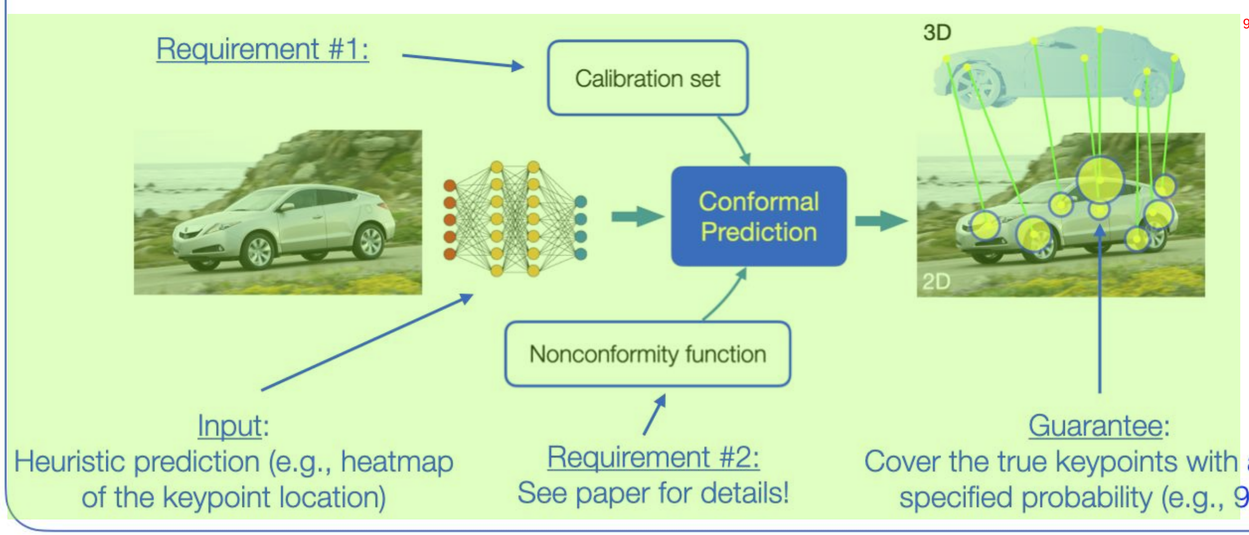


[5]

Our method: conformal keypoint detection [6]

Paradigm shift: from heuristic prediction to probabilistically correct prediction sets [7]

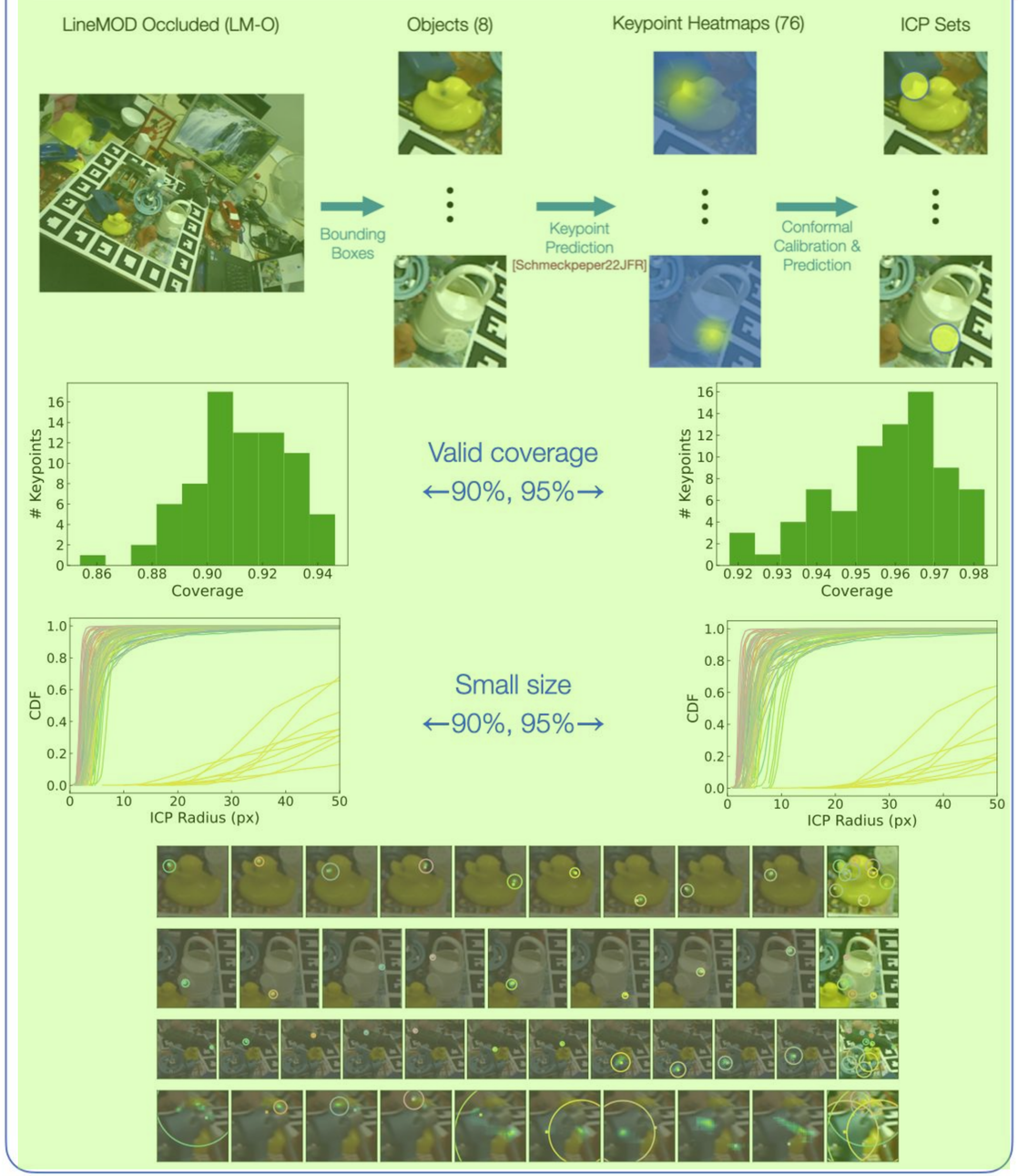
Key ingredient: inductive conformal prediction (ICP) — finite-sample uncertainty quantification without assumptions on the data distribution or the prediction model [8]



[9]

Experiments: works out of the box! [10]

Setup: 200 images for calibration, 1214 images for testing [11]

User: give me 90% and 95% coverage of the true keypoints

[12]